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One Hot Encoding

Nominal category can be converted to integer for ML models using one Hot Encoding (OHE)

Example of how it is done :

consider an dataset with 3 categories of color (we can't categorize the colors based on orders (higher/lower) so this comes under Nominal categories.

Color	Target
Red	0
Yellow	1
Blue	1
Blue	0
Red	1
Yellow	1
yellow	0

when you apply one hot encoding →

Color_R	color_Y	color_B	Target
1	0	0	0
0	1	0	1
0	0	1	1
0	0	1	0
1	0	0	1
0	1	0	1
0	1	0	0

When you apply OHE to nominal dataset, each categories form a seperate column and the values are organized in form of vector.

say If it is Red [1 0 0], yellow [0 1 0]

Blue [0 0 1]

The problem with OHE is multicollinearity.

Which means each column after OHE will have mathematical relationship between them. If we are loading these dataset into Machine learning model it will not give a accurate output.

The input (independent features) column always need to be free of mathematical relationship for accurate result.

If we see above example if we add the vector numbers, all the column will give same output as $\Sigma colors = 1$. This is called Dummy variable trap.

To solve this, we have to remove one column from the dataset, In above example n-colors = 3

~~Color-Y~~	color-B	color-R	Target

we can solve by subtracting one column $(n\text{-colors}) - 1 = 2$

Now, we have only 2 columns of colours

color_B	color_R	Target	
0	0	1	→ This means yellow
1	0	0	→ Blue
0	1	1	→ Red.

In some dataset we may have 50 (or) even more categories, so we need to create 50 independent column using OHE. But here we can reduce the number of column by keeping most frequent category. ie., we can have individual column for most frequent category and remaining can be named as other and can be made as column name "Others".

And if a dataset has many nominal categories we can first combine any two nominal features and then go for next seperately in OHE. If we want to do everything at a same time then that can be done through column transformer method